

SSILP Math

Beginner

FUNCTIONS



Learning objectives

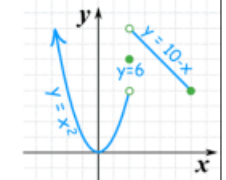
- **Understand the concept of a function** and its importance in mathematics.
 - Learn various ways to **represent functions**, including tables, graphs, and equations.
 - Identify and determine the **domain and range** of a function.
 - **Evaluate functions** for given input values.
 - Understand and work with **piecewise functions**.
- **Graph functions** and interpret their visual representations.
 - Apply the **vertical line test** to determine if a graph represents a function.
- Identify and analyze **local maxima and minima** of functions.
 - Understand the **concepts of increasing and decreasing** functions and their significance.

Session outline

1. Functions

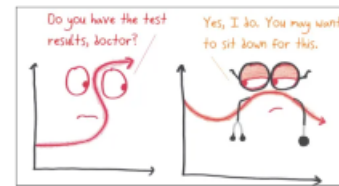
Domain and range

Piecewise functions



2. Graphs of functions

Vertical line test



Local Maxima and Local Minima



[Functions](#) (Khan academy)

(Note: Some videos may use slightly different notation & terminology. Please use the notation & terms in the prescribed textbook for this course)

1. Functions

Functions

What you need to know

Refer to the textbook!

Pre-calculus : Mathematics for Calculus
By Stewart, Redlin, Watson (7th Ed)

**Chapter 2,
Section 2.1, 2.2, 2.3**

Understand
what is a
function.

Know ways of
representing
functions.

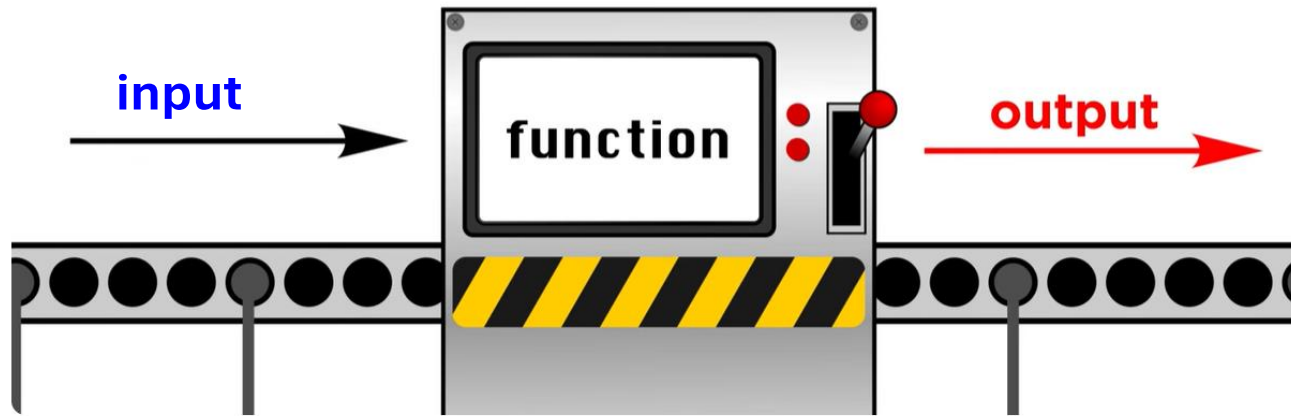
Find the **domain**
and **range** of a
function.

Evaluate
functions.

What is a function?

7

A *function* is a **rule** that describes how one quantity depends on another



We can also think of a function **like a machine**:

when we insert **input** values,
it spits out the **output** values

Examples of functions in Architectural Design

1 SITE GRADING

- Slope is at a minimum of five percent
- Slope continues for at least 10'

2 LOAD BEARING VALUE OF SOIL

- Foundation rests on firm undisturbed soil

3 LOAD BEARING VALUE OF SOIL

- Footings extend to a minimum depth of 12" below previously undisturbed soil

4 WIDTH OF FOOTINGS

- Footing width is under code

5 LEVELNESS

- Top of the footings are level
- Bottom of the footing is level (exception if building on a sloped grade)
- Spread footing is no less than 6" thick and projects on both sides, no less than 2"

6 STRENGTH OF FOOTING CONCRETE

- Foundations and slabs have a minimum of 2,500 psi concrete strength
- Walls exposed to weather meet code

Dimensions of a buildings and its components

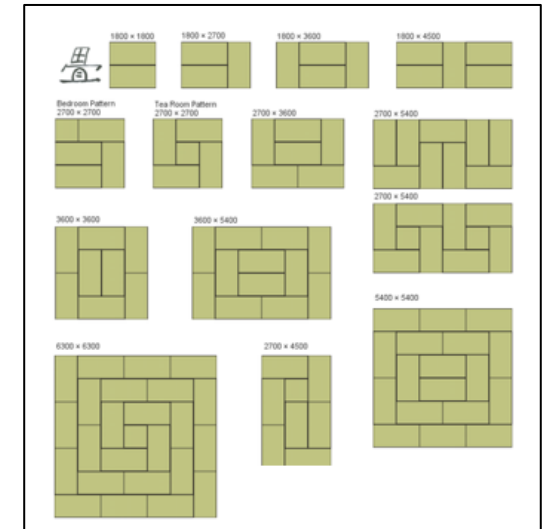
building codes, bylaws, construction standards

Building Construction Materials

S.No.	Material	Qty	Rate	Amount
1	Cement (Bags)	400	320	128000
2	Steel (Kg)	2500	45	112500
3	Sand (Cu.ft.)	1200	52	62400
4	Gravel (Cu. ft.)	1350	30	40500
5	Bricks (Nos.)	7,320	7	51240
6	Tiles (Sq. ft.)	700	40	28000
7	Color (Liters)	120	200	24000
Total Cost				4,46,640 ₹

1000 sq ft House Construction Cost

Construction cost of a building



Layout planning of buildings: E.g. Tatami Architecture

A Tatami is a type of mat which is used as a flooring material in traditional Japanese-style rooms. There are rules concerning the number of Tatami mats and the layout of the Tatami mats in a room.

How building component number and dimensions impact layout

input

Building A		name
-	10'0"	external corridor width
-	10'0"	internal corridor width
-	10'0"	staircase width
-	29'0"	staircase depth
-	6'0"	staircase offset
-	3	floors
-	4	top block units
-	2	bottom block units



Formal definition of a function

DEFINITION OF A FUNCTION

A **function** f is a rule that assigns to each element x in a set A exactly one element, called $f(x)$, in a set B .

Dependant variable [output] $y = f(\textcolor{red}{x})$ or $f : A \rightarrow B$

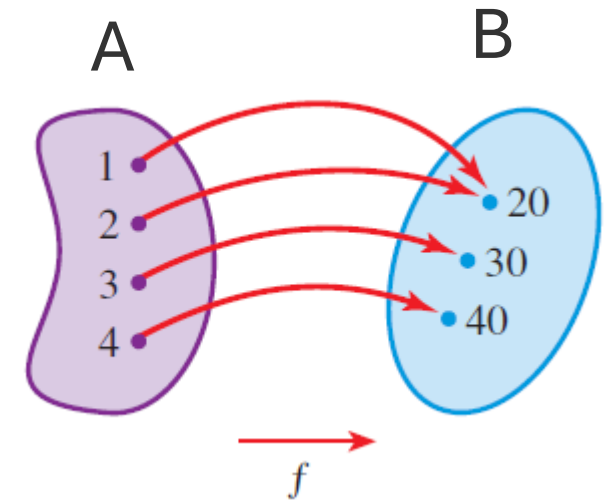
Independent variable [input]

f denotes the function name

$\textcolor{red}{x}$ is an element from the domain set **A**

y or $f(\textcolor{red}{x})$ is an element from the set **B**

The arrow indicates the mapping of input to the output



It's called **dependant** because it **depends** on the value of the **independent variable**

Function: Understanding notation

Notation

The diagram shows the notation $f(x) = x^2$. A dashed brown line on the left groups the explanatory text. Three red arrows point from the text to the corresponding parts of the equation: one from 'f is the name of the function' to the 'f', one from 'x is the input' to the 'x' inside the parentheses, and one from 'Tells you what the function does' to the 'x^2'.

$$f(x) = x^2$$

f is the name of the function x is the input Tells you what the function does

Read as
“ f of x ”

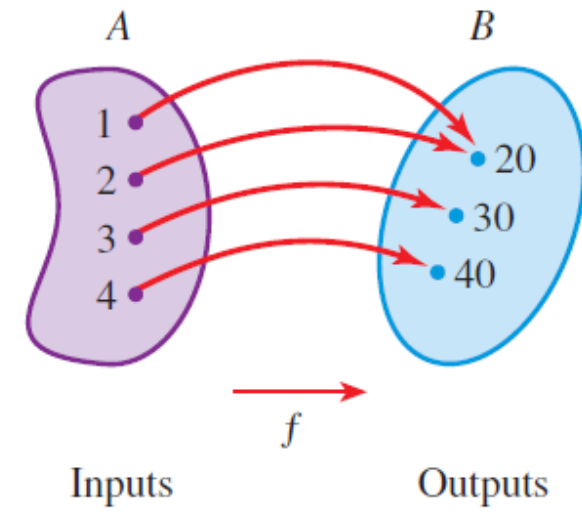
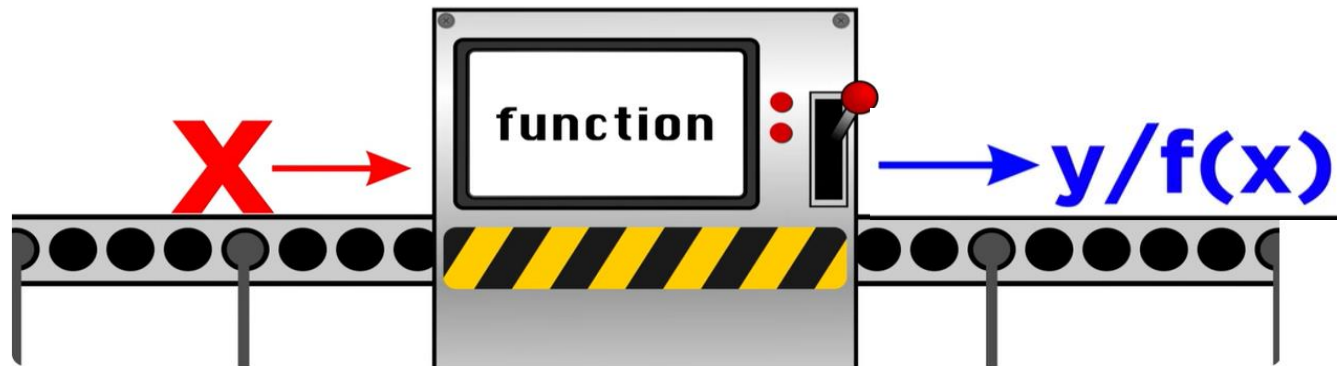
Or

“function of x ”

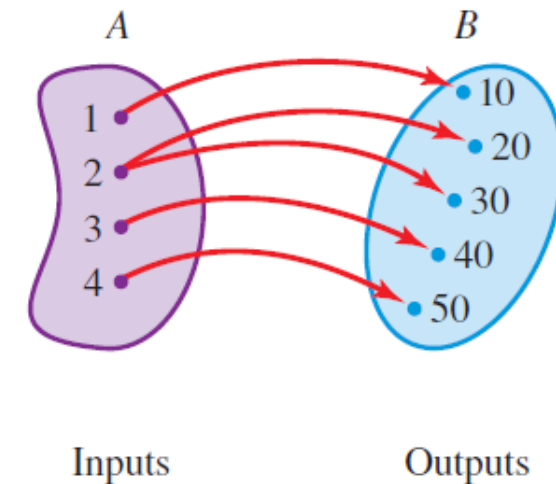
We're giving our function a name to avoid confusion with other functions. We could **use any letters** to represent functions and their input and output variables. Examples: $g(y)$, $h(u)$ or $f(t)$

Characteristic of a function

every **input** will have **one unique output**



(a) Function



(b) Not a function

Let's evaluate and represent a function

$$f(\mathbf{x}) = 0.7x$$

input
value

Plug in here!

You can make this
into a 'relation' table

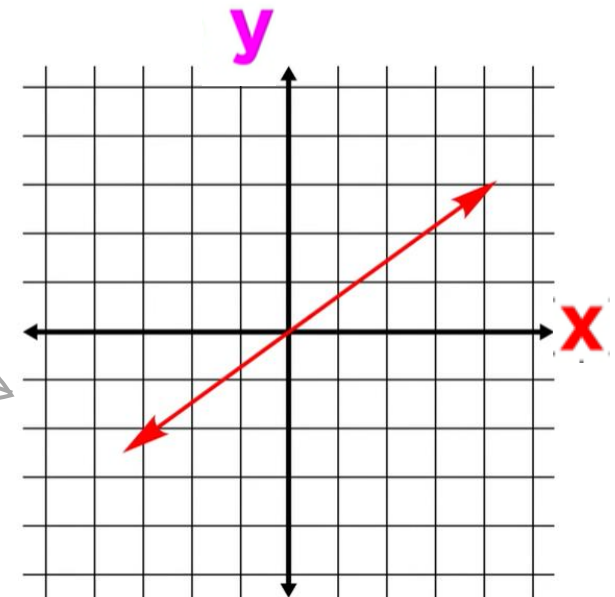
x	f(x)
1	0.7
2	1.4
3	2.1
4	2.8
5	3.5

$$f(\mathbf{1}) = 0.7(\mathbf{1}) = \mathbf{0.7}$$

$$f(\mathbf{2}) = 0.7(\mathbf{2}) = \mathbf{1.4}$$

$$f(\mathbf{3}) = 0.7(\mathbf{3}) = \mathbf{2.1}$$

Or plot it on a graph



Evaluating a function: Polynomial examples

$$f(x) = 3x^2 + x - 5$$

x acts as a **placeholder**

$$f(\square) = 3 \cdot \square^2 + \square - 5$$

To evaluate **f** at a number,
we **substitute the number**
for the **placeholder**

Example

Let $f(x) = 3x^2 + x - 5$. Evaluate each function value.

- (a) $f(-2)$ (b) $f(0)$ (c) $f(4)$ (d) $f(\frac{1}{2})$

SOLUTION To evaluate f at a number, we substitute the number for x in the definition of f .

$$(a) \quad f(-2) = 3 \cdot (-2)^2 + (-2) - 5 = 5$$

$$(b) \quad f(0) = 3 \cdot 0^2 + 0 - 5 = -5$$

$$(c) \quad f(4) = 3 \cdot (4)^2 + 4 - 5 = 47$$

$$(d) \quad f(\frac{1}{2}) = 3 \cdot (\frac{1}{2})^2 + \frac{1}{2} - 5 = -\frac{15}{4}$$

EXAMPLE

Create a table of values for the following functions:

(a) $f(x) = 2(x + 1)^2$

x	$f(x)$
-1	
0	
1	
2	
3	

(b) $g(x) = |2x + 3|$

x	$g(x)$
-3	
-2	
0	
1	
3	

ACTIVITY 1.1

1.1 Create a table of values for the following functions:

SOLUTION

1.1 Create a table of values for the following functions:

SOLUTION

Evaluate the functions at the indicated values

(a)

$$h(x) = \frac{x^2 + 4}{5};$$

$$h(2), h(-2), h(a), h(-x), h(a - 2), h(\sqrt{x})$$

(b)

$$h(t) = t + \frac{1}{t};$$

$$h(-1), h(2), h\left(\frac{1}{2}\right), h(x - 1), h\left(\frac{1}{x}\right)$$

ACTIVITY 1.2

1.2 Evaluate the functions at the indicated values

SOLUTIONS

1.2 Evaluate the functions at the indicated values

SOLUTION

Domain and range

Domain and Range

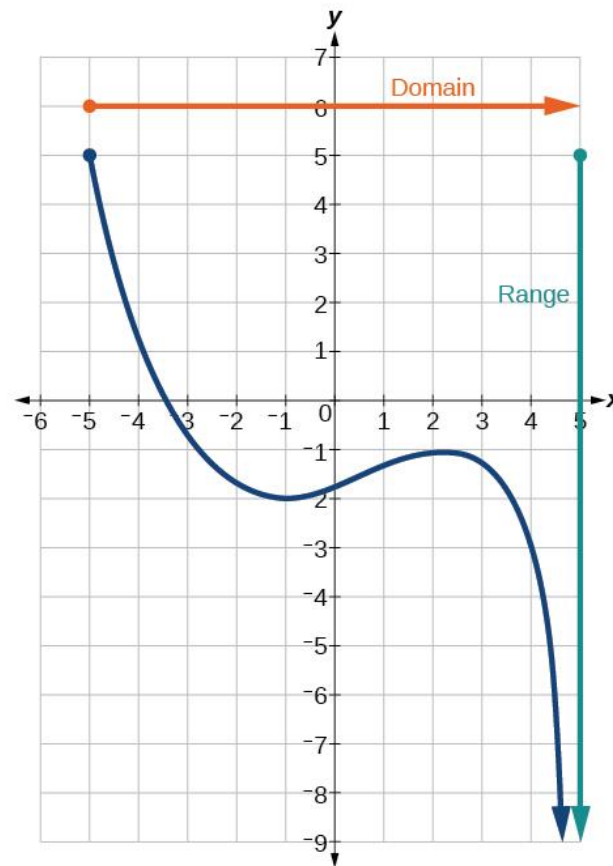
Domain

All the **possible x values** that can be plugged into the function

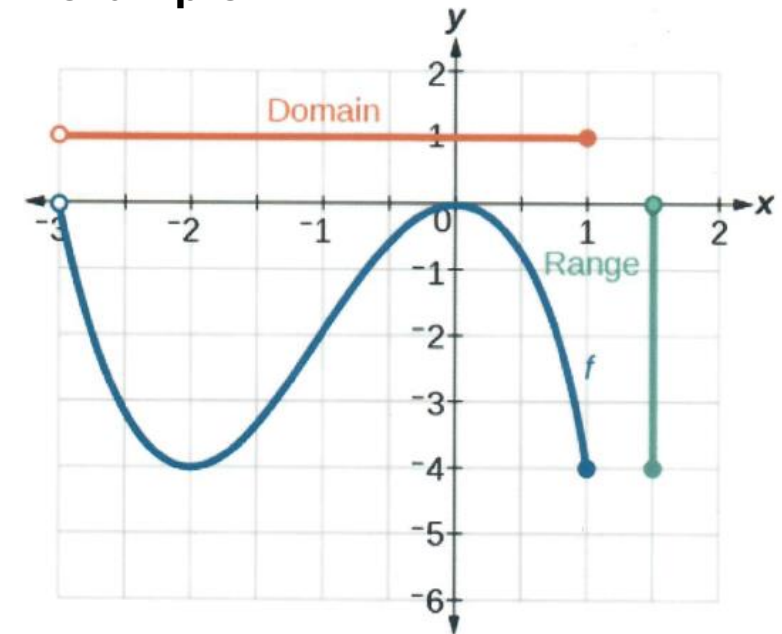
Range

All the **y values** the function can output

example



example



If the function is given by an algebraic expression and the **domain is not stated explicitly**, then by **convention** the domain of the function is the domain of the algebraic expression—that is, the **set of all real numbers** for which the expression is defined.

Domain and Range: Simple Example

Find the domain and range of the following function:

$$\{(2,10), (3,10), (4,20), (5,30), (6,40)\}.$$

Solution:

$D=\{2,3,4,5,6\}$ all the x values

$R=\{10,20,30,40\}$ all the y values

Ordered pairs!



EXAMPLE

How to find the Domain of a function

1. Identify **the input values**, usually the x .
2. Identify **any restrictions** on the input and exclude those values from the domain.
3. Write the domain in **interval form**, if possible.



Find the Domain of a Function

Find the domain of the function $f(x) = x^2 + 2$.

Solution

The input value, x , is squared and then the result is increased by two.

1. What's happening to the input value?

Any **real number** may be squared and then be increased by two, so there are **no restrictions** on the domain of this function.

2. What are the restrictions on the input?

The domain is **all real numbers**.

3. Write domain in interval form!

In interval form, the domain of f is $(-\infty, \infty)$.

EXAMPLE

Find the Domain of a Function Involving a Denominator

Find the domain of the function

$$g(x) = \frac{x+2}{x-3}$$



Denominator
cannot be zero!

Solution

The input value is x . **The function is undefined when it is divided by 0.**



Find the excluded values of x by setting the denominator equal to 0 and solving for x .

$$\begin{aligned} x - 3 &= 0 \\ x &= 3 \end{aligned}$$

Now, exclude that value, 3, from the domain. The domain is all real numbers except 3. This can be written as $x \neq 3$.

This can also be written as $x < 3$ or $x > 3$.

These can be combined using the union symbol, $\cup$.
In interval form, the domain of g is $(-\infty, 3) \cup (3, \infty)$.

EXAMPLE

Find the Domain of a Function with an Even Root

Find the domain of the function

$$h(x) = \sqrt{x + 6}.$$

Solution

The input variable is x .



Even root!
Radicand cannot
be negative

When there is an even root in the equation, all negative numbers are excluded from the radicand because even roots of negative number produce unreal results.



Set the radicand greater than or equal to zero and solve for x .

$$\begin{aligned} x + 6 &\geq 0 \\ x &\geq -6 \end{aligned}$$

The domain is all numbers greater than or equal to -6 .

In interval form, the domain of h is $[-6, \infty)$.

EXAMPLE

Analyze a function: Evaluate, find domain and range

A function f is defined by the formula

$$f(x) = x^2 + 4$$

- (a) Express in words how f acts on the input x to produce the output $f(x)$.
- (b) Evaluate $f(3)$, $f(-2)$, and $f(\sqrt{5})$.
- (c) Find the domain and range of f .

SOLUTION

- (a) The formula tells us that f first squares the input x and then adds 4 to the result. So f is the function

“square, then add 4”

- (b) The values of f are found by substituting for x in the formula $f(x) = x^2 + 4$.

$$f(3) = 3^2 + 4 = 13 \quad \text{Replace } x \text{ by } 3$$

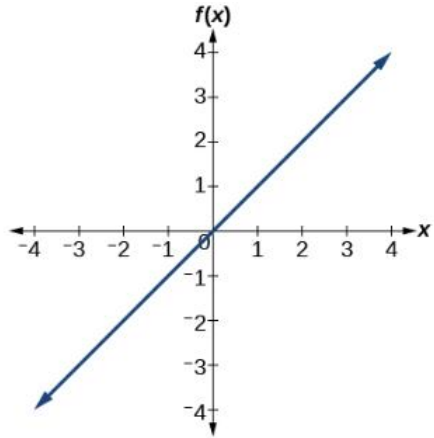
$$f(-2) = (-2)^2 + 4 = 8 \quad \text{Replace } x \text{ by } -2$$

$$f(\sqrt{5}) = (\sqrt{5})^2 + 4 = 9 \quad \text{Replace } x \text{ by } \sqrt{5}$$

- (c) The domain of f consists of all possible inputs for f . Since we can evaluate the formula $f(x) = x^2 + 4$ for every real number x , the domain of f is the set $\mathbb{R}$ of all real numbers.

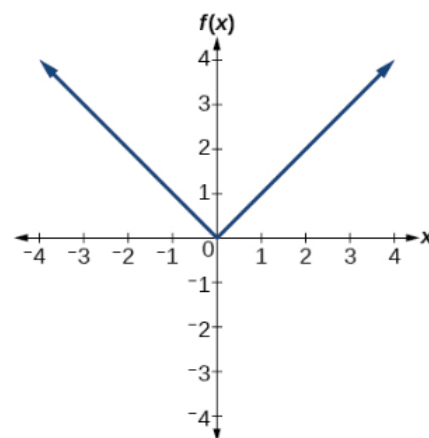
The range of f consists of all possible outputs of f . Because $x^2 \geq 0$ for all real numbers x , we have $x^2 + 4 \geq 4$, so for every output of f we have $f(x) \geq 4$. Thus the range of f is $\{y \mid y \geq 4\} = [4, \infty)$.

Domain and range of 'toolkit' functions



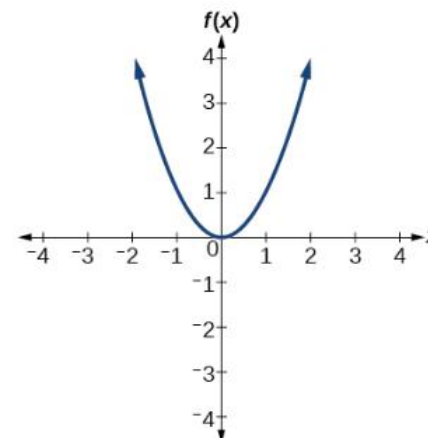
Domain: $(-\infty, \infty)$
Range: $(-\infty, \infty)$

Identity function $f(x) = x$



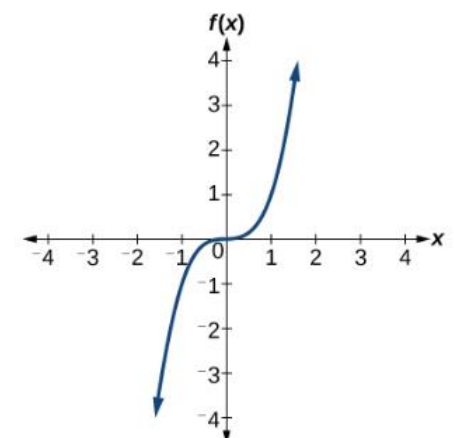
Domain: $(-\infty, \infty)$
Range: $[0, \infty)$

Absolute function $f(x) = |x|$



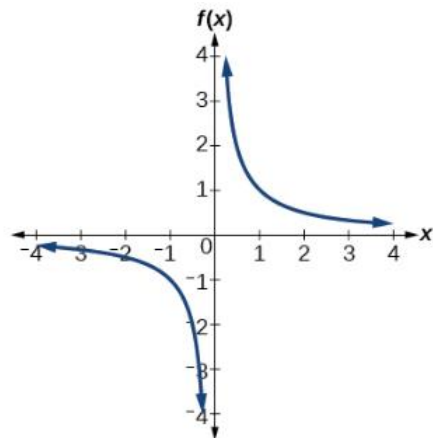
Domain: $(-\infty, \infty)$
Range: $[0, \infty)$

Quadratic function $f(x) = x^2$



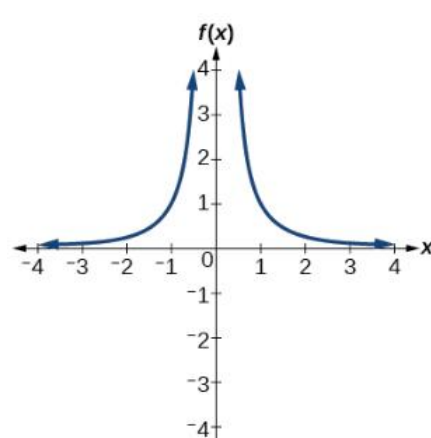
Domain: $(-\infty, \infty)$
Range: $(-\infty, \infty)$

Cubic function $f(x) = x^3$



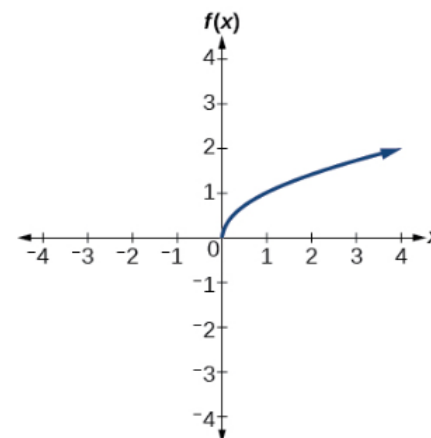
Domain: $(-\infty, 0) \cup (0, \infty)$
Range: $(-\infty, 0) \cup (0, \infty)$

Reciprocal function $f(x) = \frac{1}{x}$



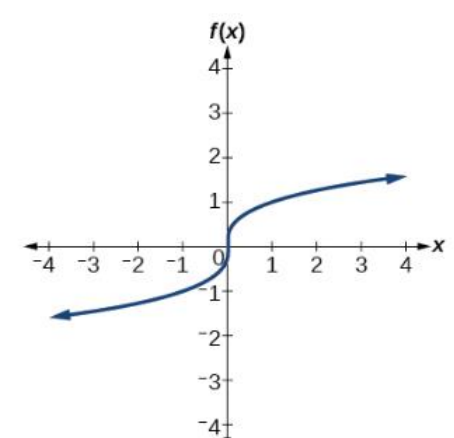
Domain: $(-\infty, 0) \cup (0, \infty)$
Range: $(0, \infty)$

Reciprocal squared function $f(x) = \frac{1}{x^2}$



Domain: $[0, \infty)$
Range: $[0, \infty)$

Square root function $f(x) = \sqrt{x}$



Domain: $(-\infty, \infty)$
Range: $(-\infty, \infty)$

Cube root function $f(x) = \sqrt[3]{x}$

Find the domain and range of the function

(a) $f(x) = 5x^2 + 4$

(b) $f(x) = 5x^2 + 4, \quad 0 \leq x \leq 2$

1.3 Find the domain and range of the function

SOLUTION

1.3 Find the domain and range of the function

SOLUTION

Find the domain of the function

(a) $f(x) = \frac{1}{3x - 6}$

(b) $g(t) = \sqrt{t^2 + 9}$

(c) $g(x) = \sqrt{7 - 3x}$

(d) $f(x) = \frac{x^2}{\sqrt{6 - x}}$

ACTIVITY 1.4

1.4 Find the domain of the function

SOLUTION

1.4 Find the domain of the function

SOLUTION

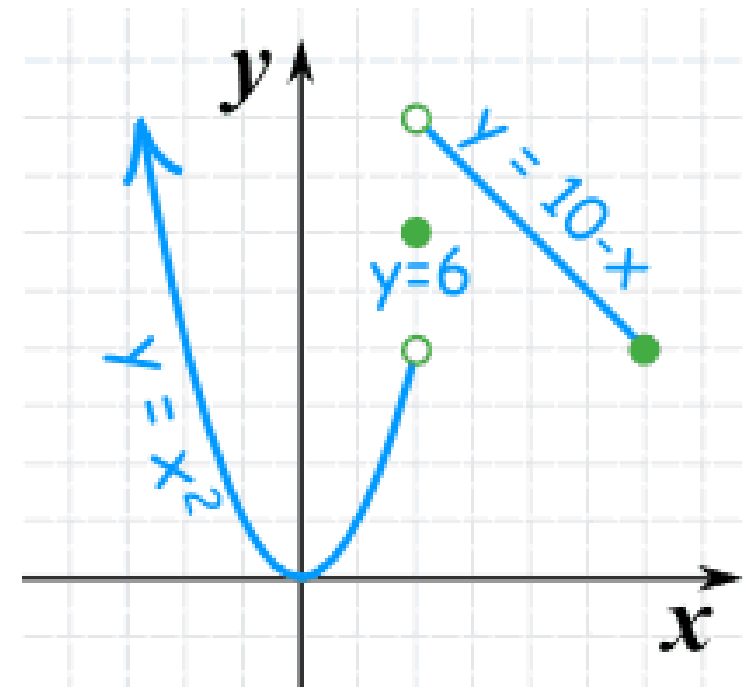
1.4 Find the domain of the function

SOLUTION

1.4 Find the domain of the function

SOLUTION

Piecewise functions



Piecewise Functions

A **piecewise function** is a function in which more than one formula is used to define the output over different pieces of the domain.

Piecewise functions are used to describe situations where **the rule or relation changes as the input crosses certain boundaries**.

General form

$$f(x) = \begin{cases} \text{formula 1, if domain 1} \\ \text{formula 2, if domain 2} \\ \text{formula 3, if domain 3} \end{cases}$$

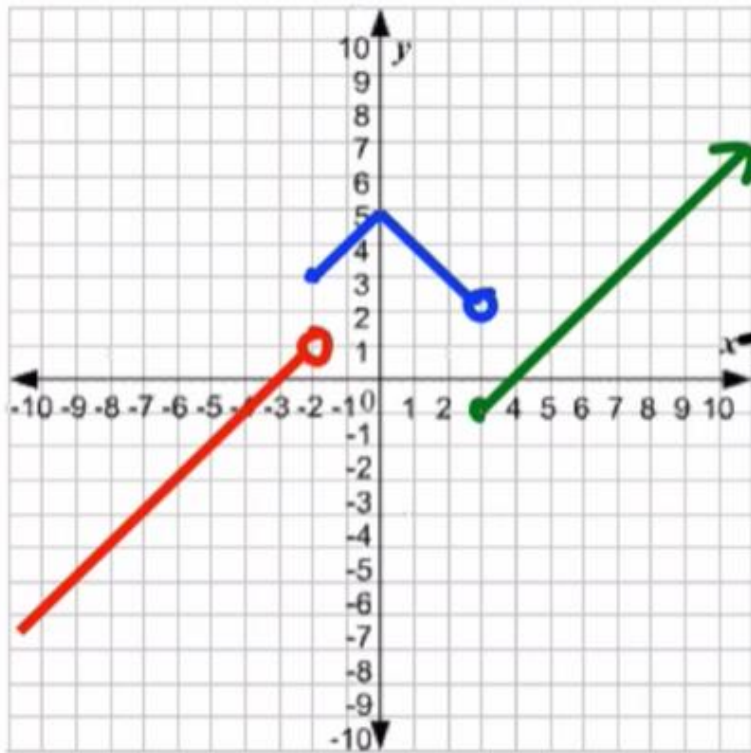
Example

$$p(x) = \begin{cases} 0.55, & \text{if } x \leq 1 \\ 0.75, & \text{if } 1 < x \leq 2 \\ 0.95, & \text{if } 2 < x \leq 3 \\ 1.15, & \text{if } 3 < x \leq 4 \\ \vdots & \end{cases}$$

Made up of **several different equations** or **pieces**.

Each piece is **evaluated differently**

Piecewise functions: Example



$$f(x) = \begin{cases} x+3 & \text{if } x < -2 \\ -|x|+5 & \text{if } -2 \leq x < 3 \\ x-4 & \text{if } x \geq 3 \end{cases}$$

EXAMPLE

Summary: Four ways to represent a function

FOUR WAYS TO REPRESENT A FUNCTION

Verbal Using words:

“To convert from Celsius to Fahrenheit, multiply the Celsius temperature by $\frac{9}{5}$, then add 32.”

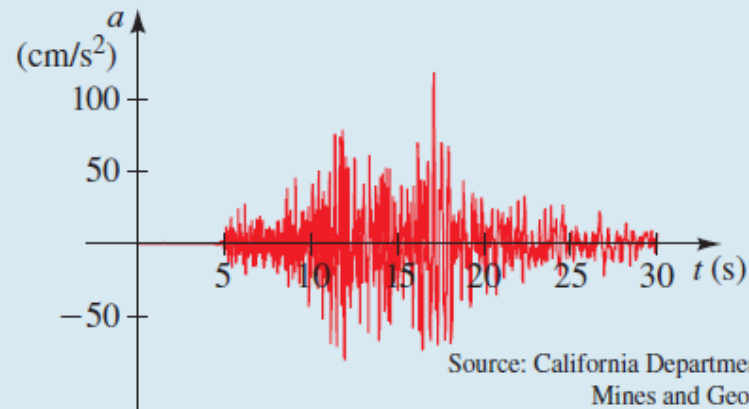
Relation between Celsius and Fahrenheit temperature scales

Algebraic Using a formula:

$$A(r) = \pi r^2$$

Area of a circle

Visual Using a graph:



Source: California Department of Mines and Geology

Vertical acceleration during an earthquake

Numerical Using a table of values:

w (ounces)	$C(w)$ (dollars)
$0 < w \leq 1$	\$0.98
$1 < w \leq 2$	\$1.19
$2 < w \leq 3$	\$1.40
$3 < w \leq 4$	\$1.61
$4 < w \leq 5$	\$1.82
$\vdots$	$\vdots$

Cost of mailing a large first-class envelope

Evaluate the Piecewise Defined Functions

(a)

$$f(x) = \begin{cases} 5 & \text{if } x \leq 2 \\ 2x - 3 & \text{if } x > 2 \end{cases}$$

$$f(-3), f(0), f(2), f(3), f(5)$$

(b)

$$f(x) = \begin{cases} x^2 + 2x & \text{if } x \leq -1 \\ x & \text{if } -1 < x \leq 1 \\ -1 & \text{if } x > 1 \end{cases}$$

$$f(-4), f(-\frac{3}{2}), f(-1), f(0), f(25)$$

ACTIVITY 1.5

1.5 Evaluate the Piecewise Defined Functions

SOLUTIONS

1.5 Evaluate the Piecewise Defined Functions

SOLUTIONS

2. Graphs of functions

What are graphs?

We can represent a function using a graph.

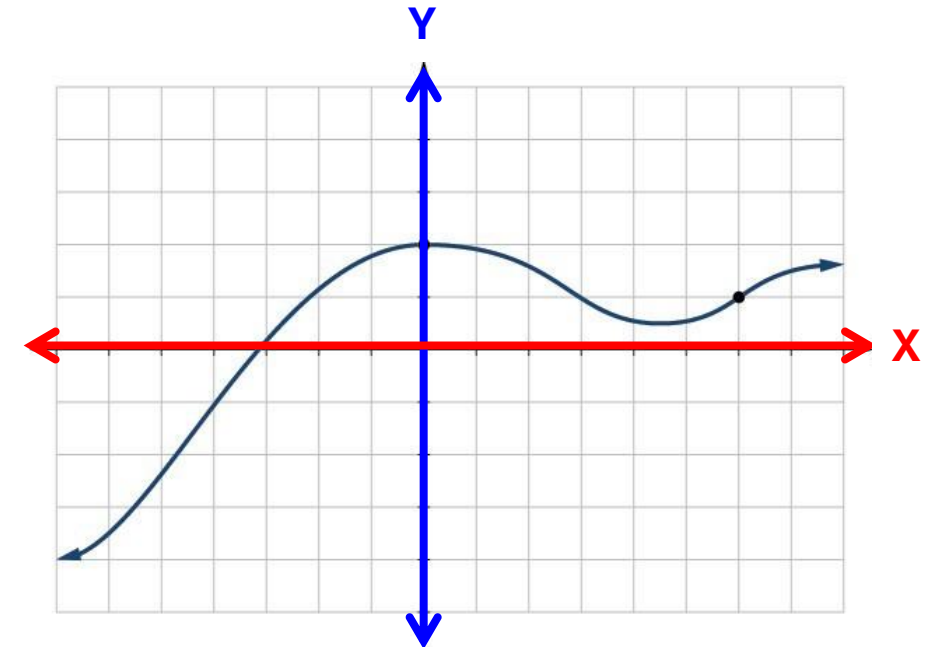
“ordered pairs”

Graphs display many **input-output pairs** in a small space.

The visual information they provide often makes relationships easier to understand.

We typically construct graphs with the

- **input values** along the horizontal axis and the
- **output values** along the vertical axis.



Interval notation vs coordinates notation

Sometimes certain Interval Notation and coordinates of a point look the same, which can be confusing. However, **in context you should be able to tell** whether you are looking at Interval Notation or the coordinates of a point.

- For example, in **Interval Notation, $(2, 6)$ would indicate an interval** along the x-axis between 2 and 6, but not including 2 and 6.
- **As coordinates, $(2, 6)$ would indicate the location of a point** in the Cartesian Coordinate System with x-coordinate 2 and y-coordinate 6.

How to make a table of values

Make a table of values for
 $f(x) = 3x + 2$.

1

Make a two-column table. Label the columns x and $f(x)$.

x	$f(x)$

2

Choose several values for x and put them as separate rows in the x column.

Tip: It's always good to include 0, positive values, and negative values, if you can.

x	$f(x)$
-2	
-1	
0	
1	
3	

3

Evaluate the function for each value of x , and write the result in the $f(x)$ column next to the x value you used.

When $x = 0$, $f(0) = 3(0) + 2 = 2$, $f(1) = 3(1) + 2 = 5$,
 $f(-1) = 3(-1) + 2 = -3 + 2 = -1$ and so on.

x	$f(x)$
-2	-4
-1	-1
0	2
1	5
3	11

4

(Note that your table of values may be different from someone else's. You may each choose different numbers for x .)

x	$f(x)$
-2	-4
-1	-1
0	2
1	5
3	11

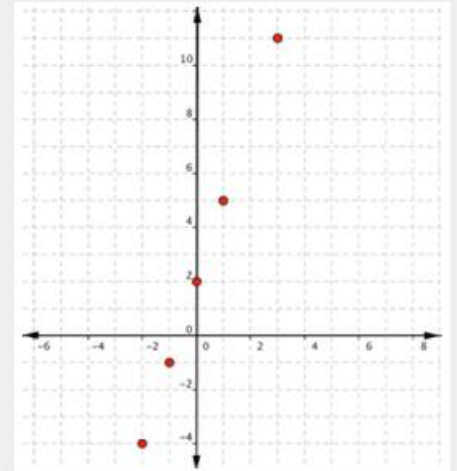
How to plot points on a graph

- 1 Start with a table of values, like the one in the previous example.
If you think of $f(x)$ as y , each row forms an ordered pair that you can plot on a coordinate grid.

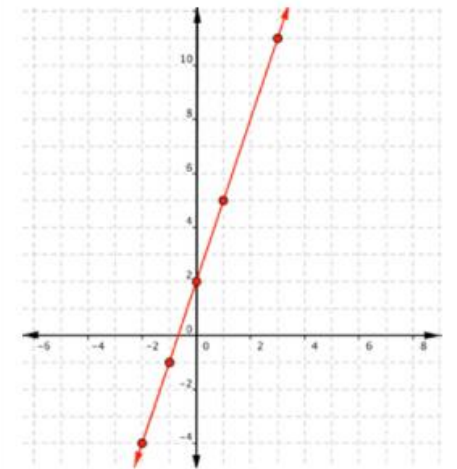
x	$f(x)$
-2	-4
-1	-1
0	2
1	5
3	11

Graph
 $f(x)=3x+2.$

- 2 Plot the points on a coordinate grid



- 3 Connect the points!



Graphing functions by plotting points

We first make a table of values.

Then we plot the points given by the table and join them by a smooth curve to obtain the graph.

$$f(x) = x^2$$

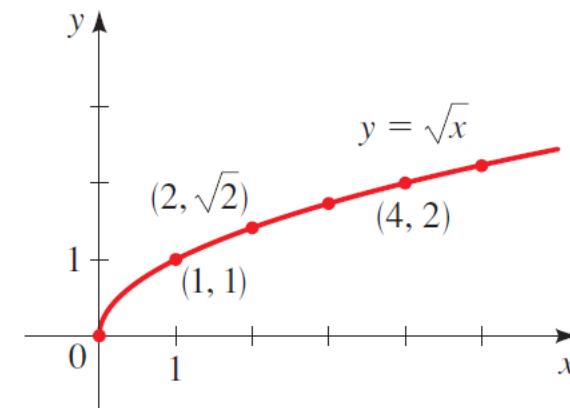
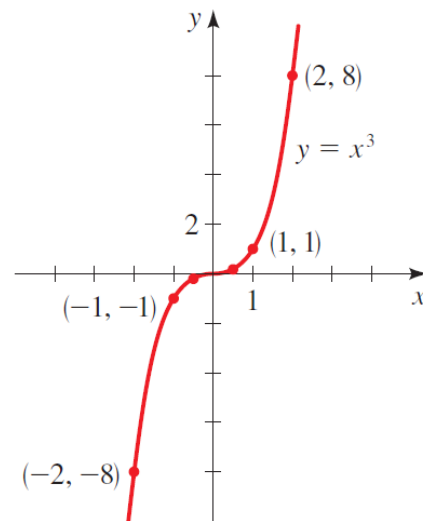
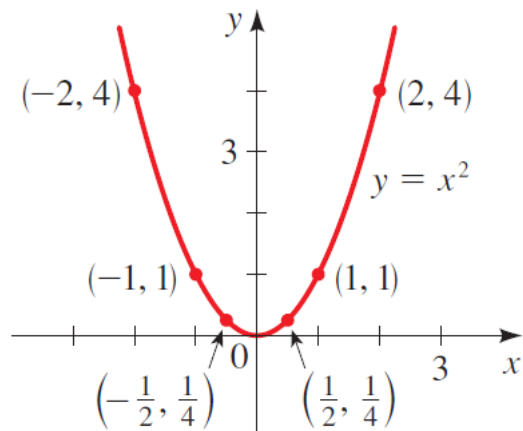
x	$f(x) = x^2$
0	0
$\pm\frac{1}{2}$	$\frac{1}{4}$
± 1	1
± 2	4
± 3	9

$$g(x) = x^3$$

x	$g(x) = x^3$
0	0
$\frac{1}{2}$	$\frac{1}{8}$
1	1
2	8
$-\frac{1}{2}$	$-\frac{1}{8}$
-1	-1
-2	-8

$$h(x) = \sqrt{x}$$

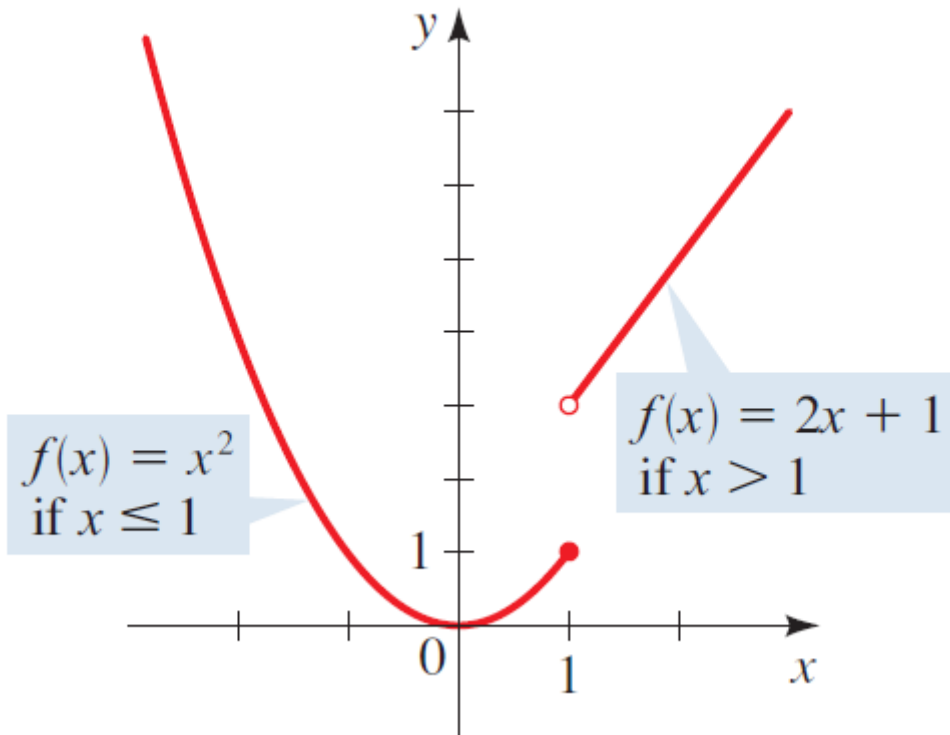
x	$h(x) = \sqrt{x}$
0	0
1	1
2	$\sqrt{2}$
3	$\sqrt{3}$
4	2
5	$\sqrt{5}$



EXAMPLE

Graphing piecewise defined functions

$$f(x) = \begin{cases} x^2 & \text{if } x \leq 1 \\ 2x + 1 & \text{if } x > 1 \end{cases}$$



The **solid dot** at (1, 1) indicates that this point is **included** in the graph

the **open dot** at (1, 3) indicates that this point is **excluded** from the graph.

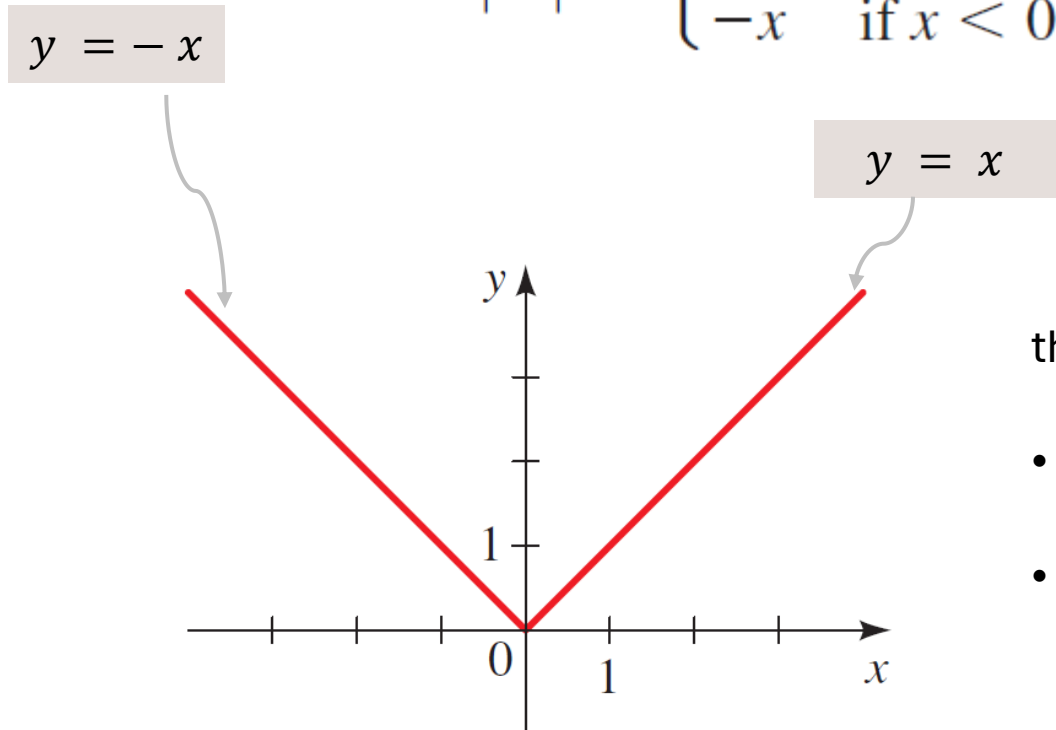
Graph of absolute value functions

Sketch a graph of the absolute value function $f(x) = |x|$

Recall that

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

Definition of absolute value!
We discussed this in week 1!



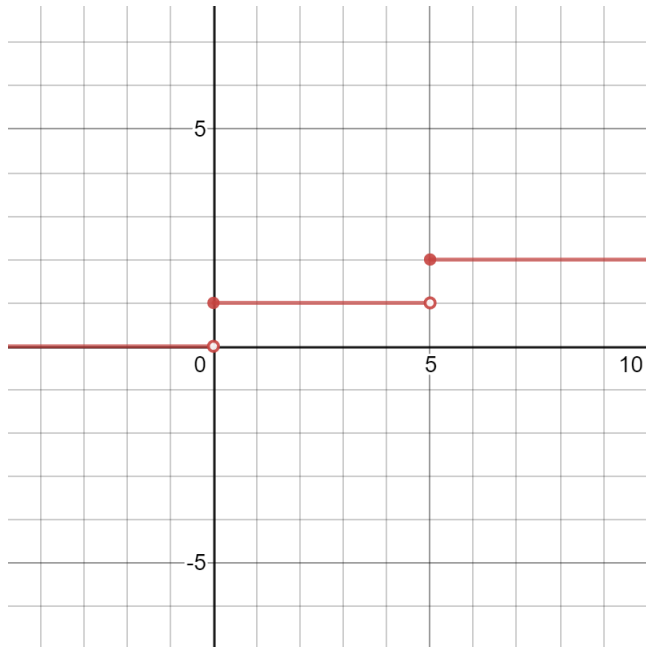
the graph of f

- coincides with the line $y = x$ to the right of the y -axis and
- coincides with the line $y = -x$ to the left of the y -axis

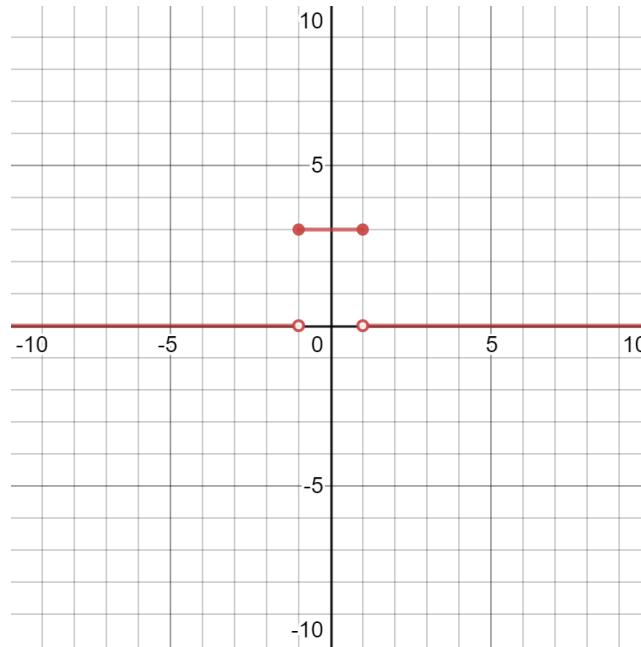
Graphs of step functions

A **step function** is a **piecewise function** over the real numbers that is made up from a **finite number of constant functions**.

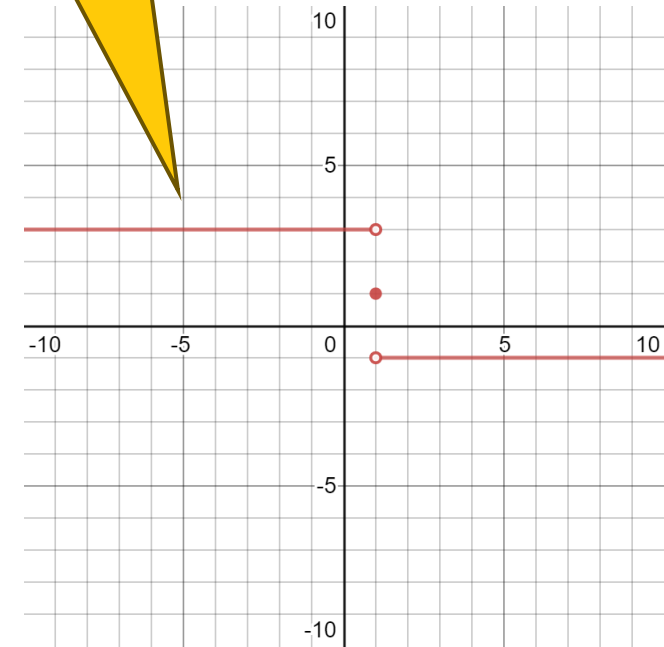
Made up of
horizontal line
segments,
Looks like a series
of steps



$$f(x) = \begin{cases} 0 & x < 0 \\ 1 & 0 \leq x < 5 \\ 2 & x \geq 5 \end{cases}$$



$$g(x) = \begin{cases} 0 & x < -1 \\ 3 & -1 \leq x \leq 1 \\ 0 & x > 1 \end{cases}$$



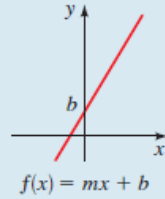
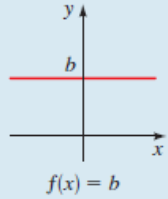
$$h(x) = \begin{cases} 3 & x < 1 \\ 1 & x = 1 \\ -1 & x > 1 \end{cases}$$

Some functions and their graphs

SOME FUNCTIONS AND THEIR GRAPHS

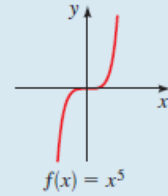
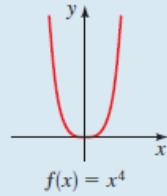
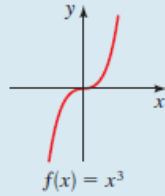
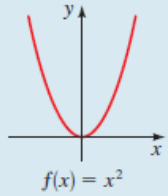
Linear functions

$$f(x) = mx + b$$



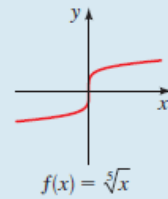
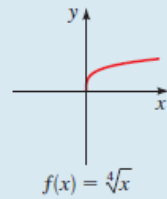
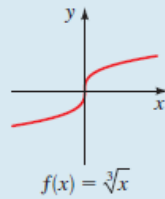
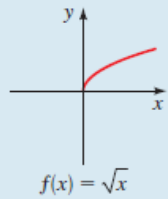
Power functions

$$f(x) = x^n$$



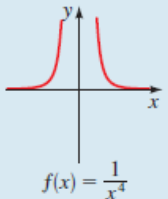
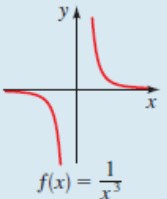
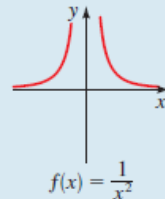
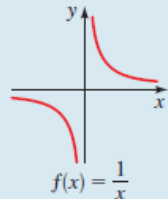
Root functions

$$f(x) = \sqrt[n]{x}$$



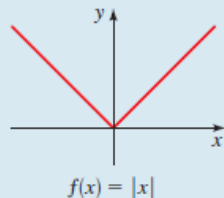
Reciprocal functions

$$f(x) = \frac{1}{x^n}$$



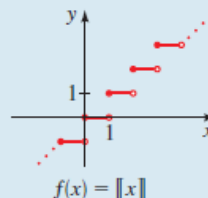
Absolute value function

$$f(x) = |x|$$



Greatest integer function

$$f(x) = \llbracket x \rrbracket$$



Useful to know the basic shape of the graphs of functions!

(a) $f(x) = -x + 3, \quad -3 \leq x \leq 3$

(b) $f(x) = -x^2$

(c) $C(t) = \frac{1}{t^2}$

2.1 Graphing Functions

2.1 Graphing Functions

2.1 Graphing Functions

Activity: Graph a Piece-wise function

$$f(x) = \begin{cases} x^2, & \text{if } x \leq 1 \\ 3, & \text{if } 1 < x \leq 2 \\ x, & \text{if } x > 2 \end{cases}$$

ACTIVITY 2.2

2.2 Graphing activity: Solution

2.3 Evaluating and graphing a piecewise function

A cell phone company uses the function below to determine the cost, C , in for g gigabytes of data transfer.

$$C(g) = \begin{cases} 25 & \text{if } 0 < g < 2 \\ 25 + 10(g - 2) & \text{if } g \geq 2 \end{cases}$$

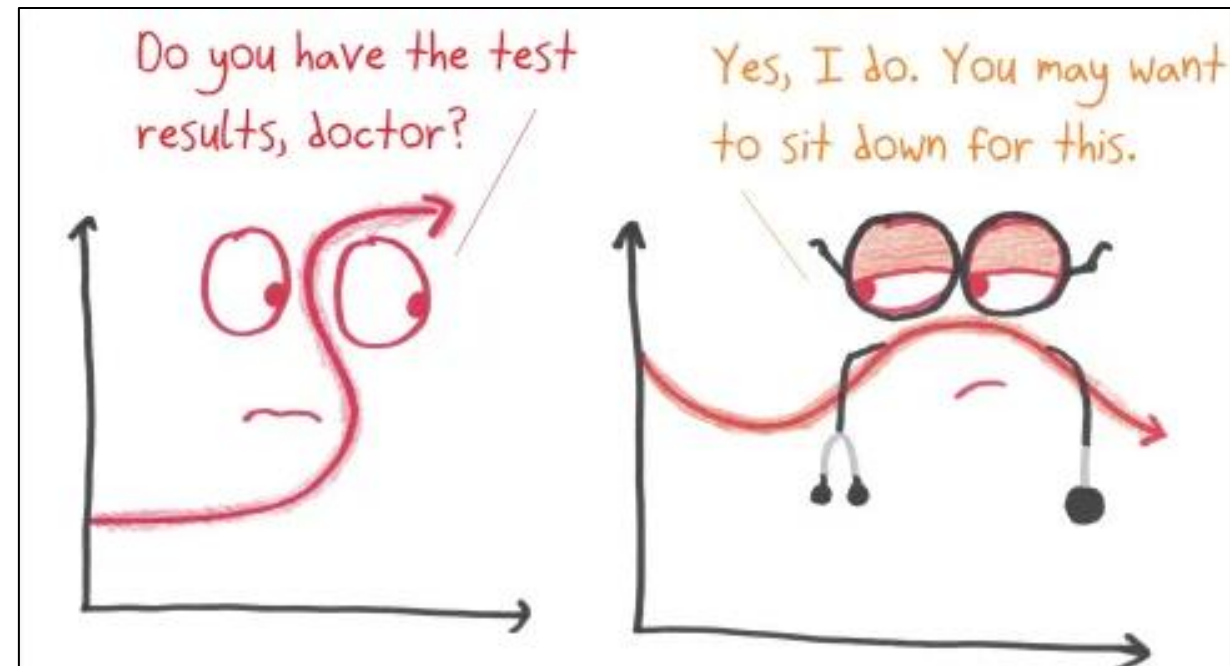
Find the cost of using 1.5 gigabytes of data and the cost of using 4 gigabytes of data. Graph the piecewise function.

ACTIVITY 2.3

Evaluating and graphing a piecewise function

SOLUTIONS

Vertical line test



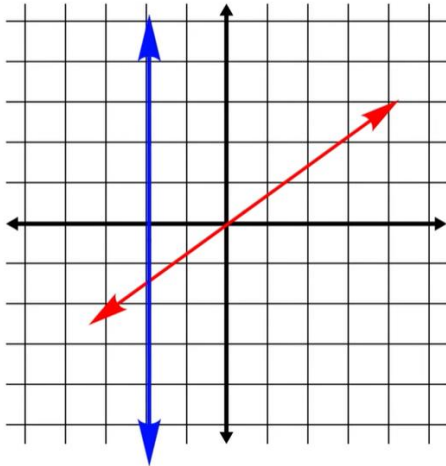
Vertical line test

THE VERTICAL LINE TEST

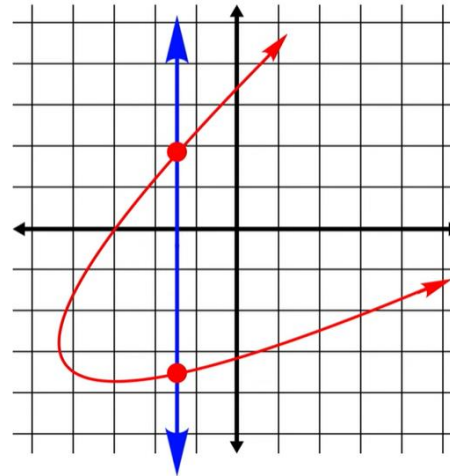
A curve in the coordinate plane is the graph of a function if and only if no vertical line intersects the curve more than once.

functions

any $x \rightarrow$ one y



a vertical line may
only intersect
the function at
one point at
any given time

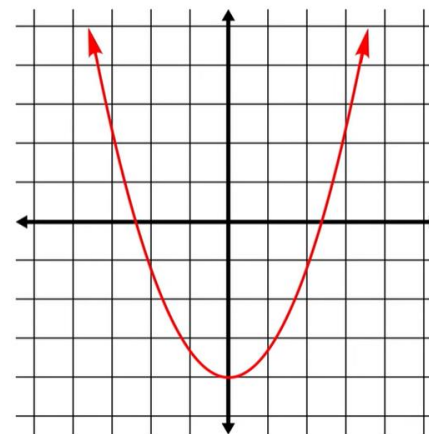
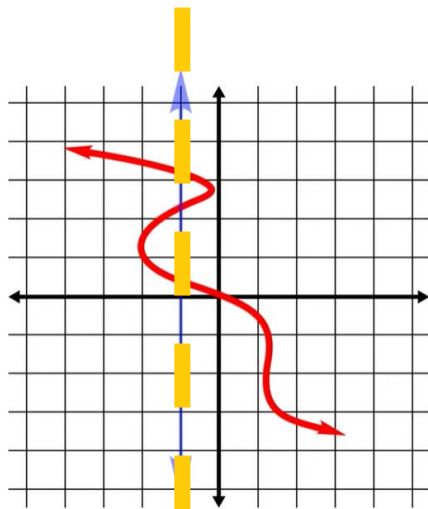
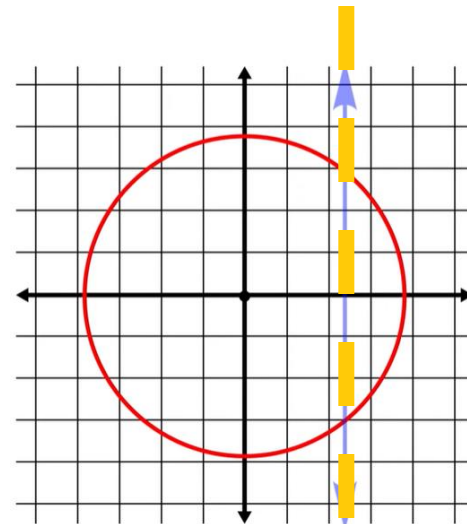
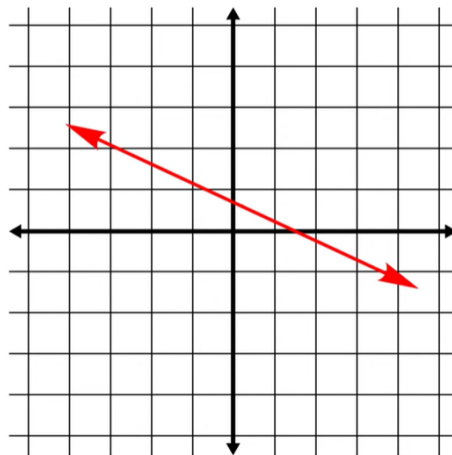
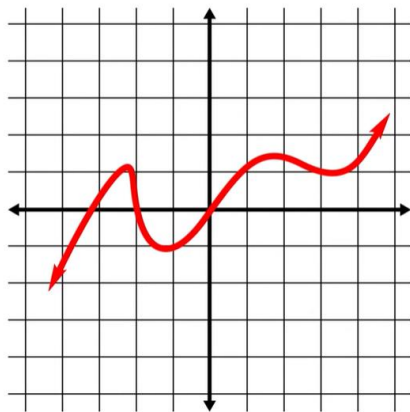


if a vertical line
intersects at **two**
or more spots
it is not
a function

If you had a function that
represented the
area of a square room
(**output**)
given a particular length
(**input**),

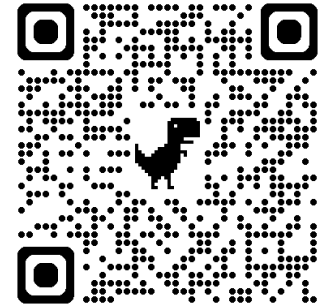
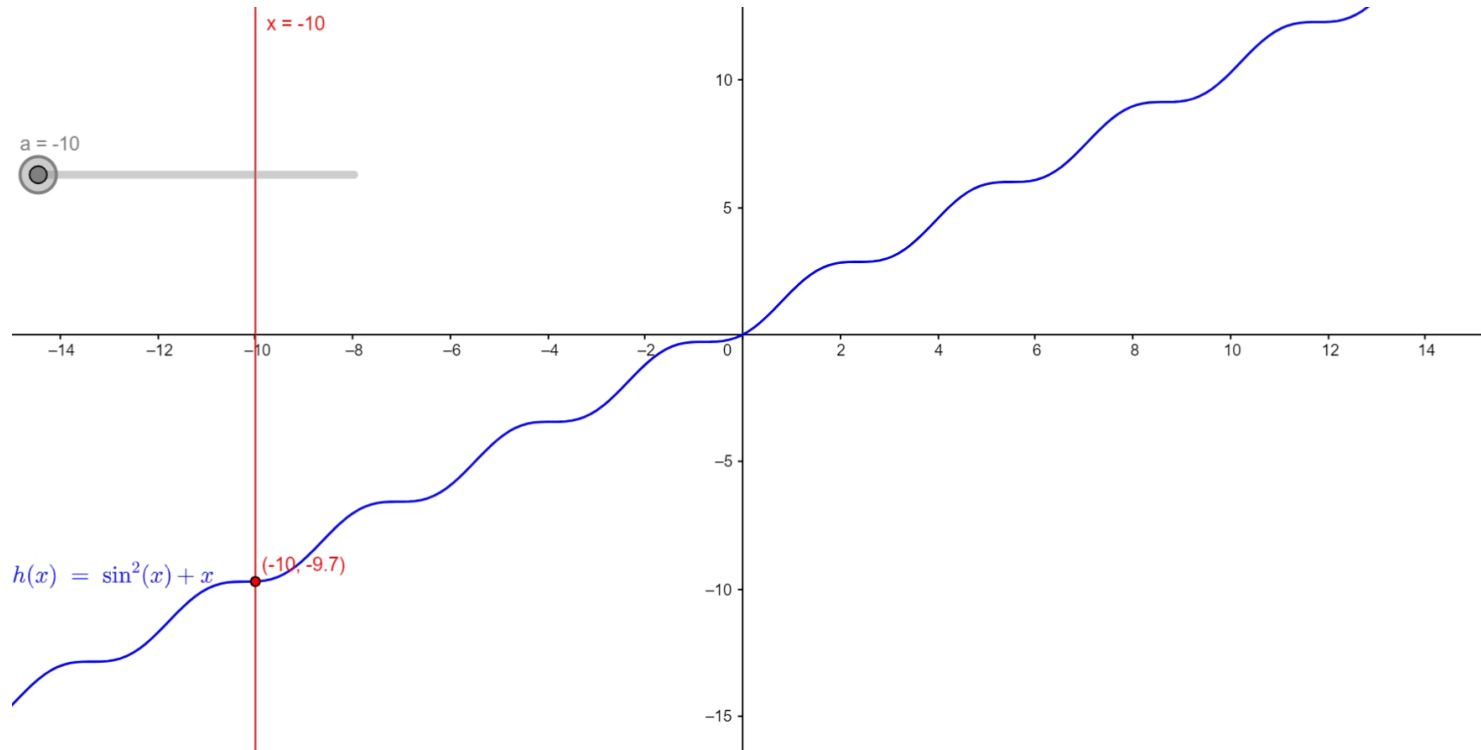
is it possible to get two
different areas for the same
room length?

Determine which of these is a function?



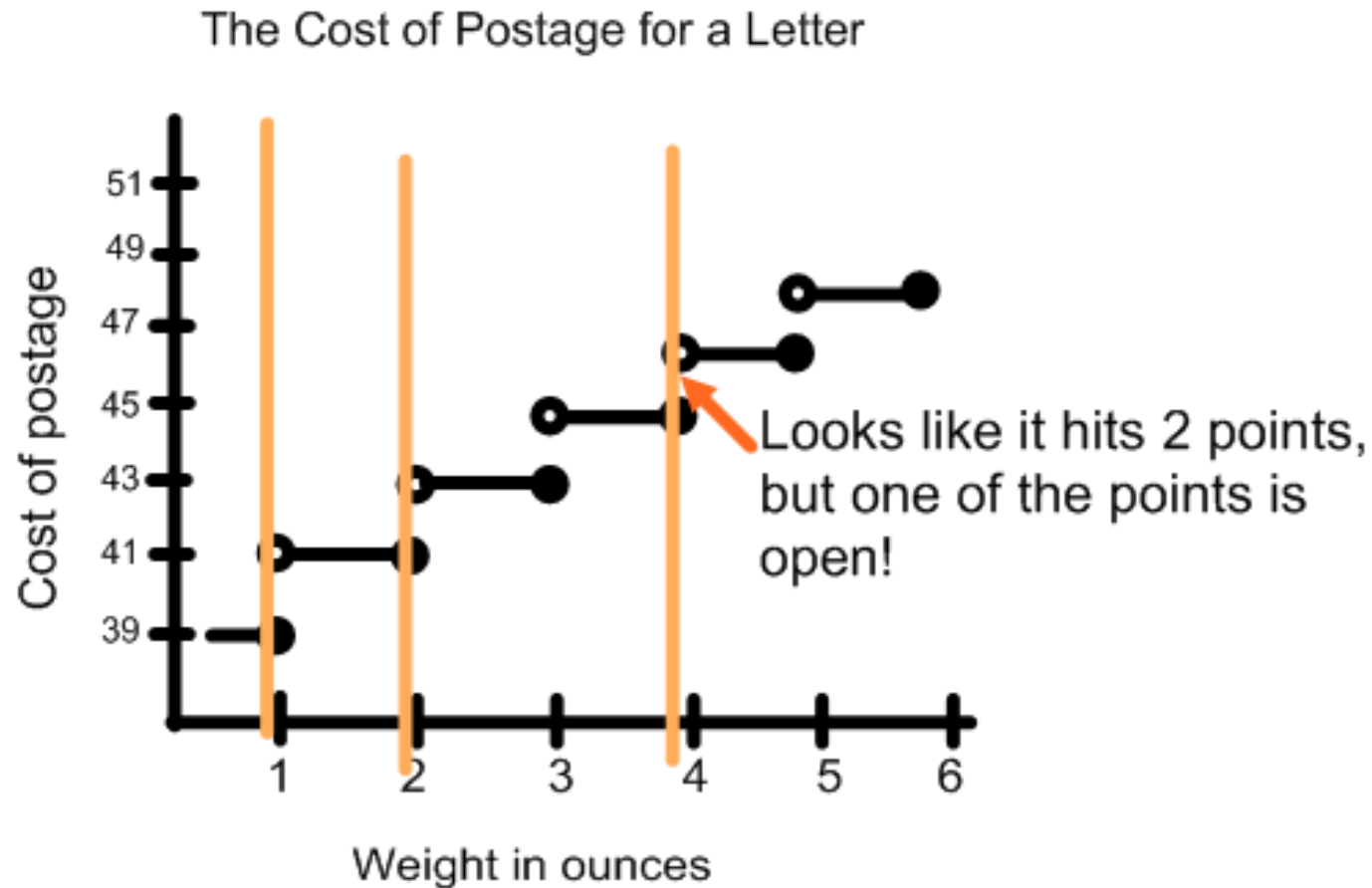
EXAMPLE

Vertical line test: Geogebra app



Graphs of step functions

Do step functions pass the vertical line test?



Remember:

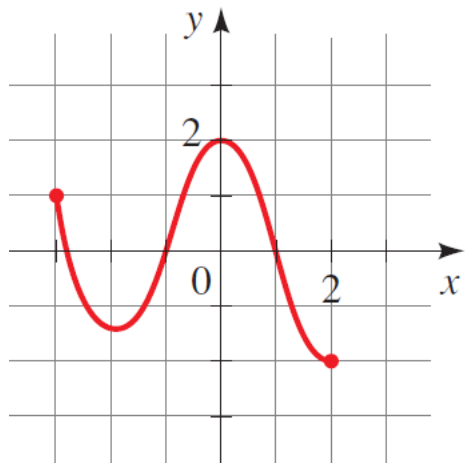
Closed circle means that the point is included.

An open circle does NOT include that point.

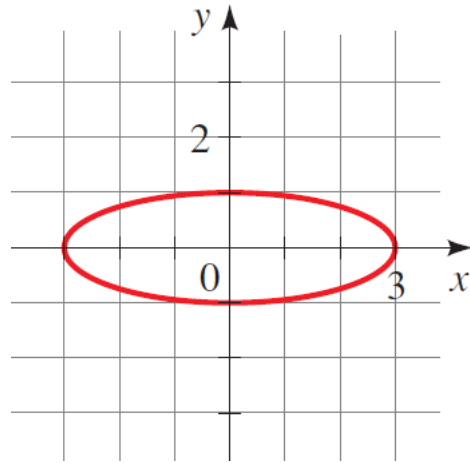
So, in this case, where it looks like the vertical line is touching two points, it is really only touching one point, because the open circle does not include that point.

Vertical Line Test: Domain and Range

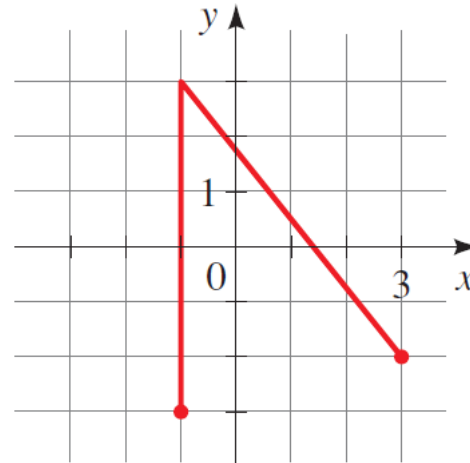
Use the Vertical Line Test to determine whether the curve is a graph of a function of x . If it is, state the domain and range of the function.



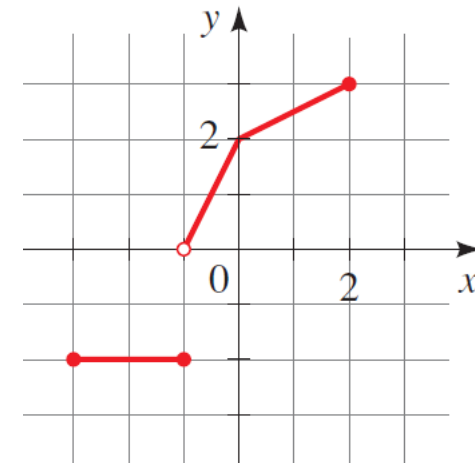
(a)



(b)



(c)



(d)

ACTIVITY 2.4

2.4 Vertical Line Test: Domain and Range

SOLUTIONS

2.4 Vertical Line Test: Domain and Range

SOLUTIONS

2.4 Vertical Line Test: Domain and Range

SOLUTIONS

2.4 Vertical Line Test: Domain and Range

SOLUTIONS

Local Maxima and Local Minima



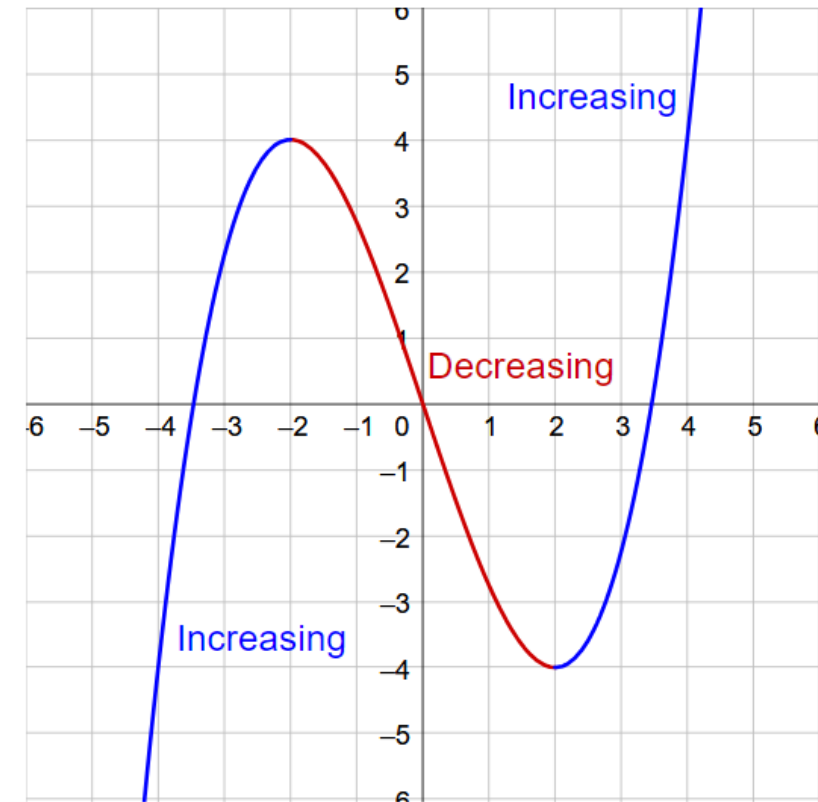
Increasing and decreasing functions

Graphs show intervals where the functions change in specific ways.

A function is **increasing** if it **rises from left to right** and has a positive rate of change.

A function is **decreasing** if it **falls from left to right** and has a negative rate of change.

A function can also be **constant** if it is horizontal and **does not rise or fall**.



The function is increasing on $(-\infty, -2) \cup (2, \infty)$ and is decreasing on $(-2, 2)$.

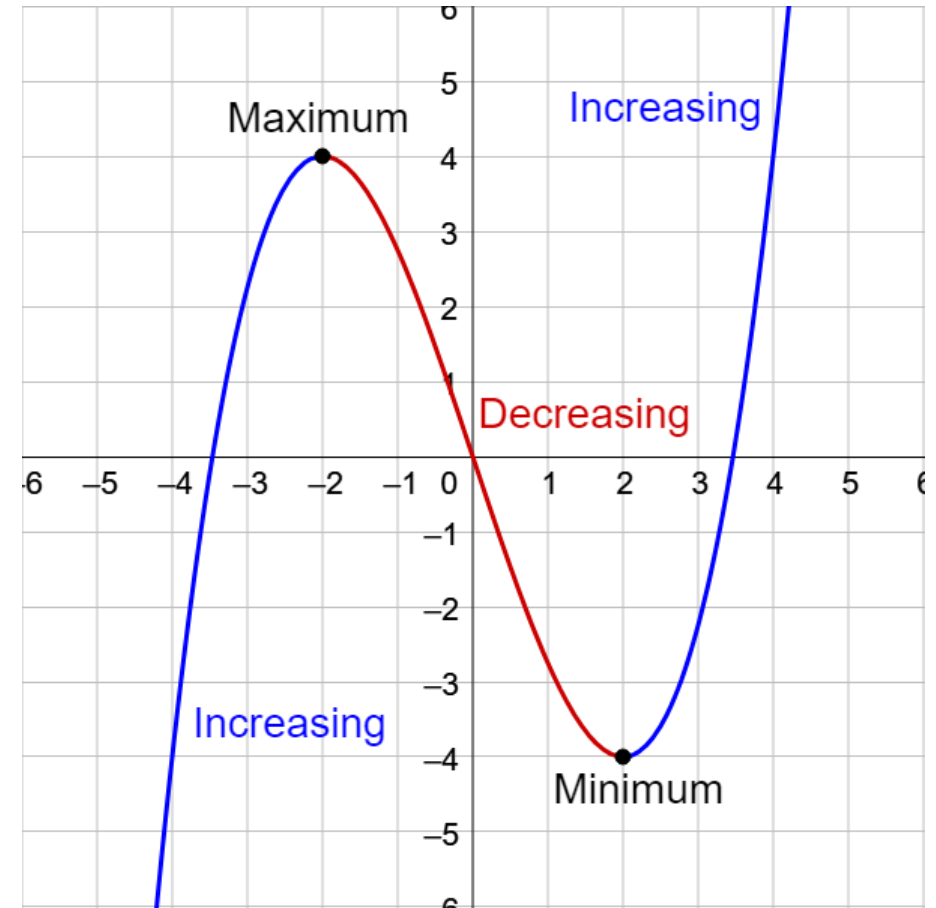
Local maximum and Local minimum

When functions change from **increasing to decreasing**, it produces the **highest point in the area** called a **local maximum**.

When a function changes from **decreasing to increasing**, it produces the **lowest point in the area** called a **local minimum**.



Local maximums and minimums are called **local extrema**



Local maximum is 4, and it occurs at $x = -2$.
Local minimum is -4 and it occurs at $x = 2$.

Activity

???



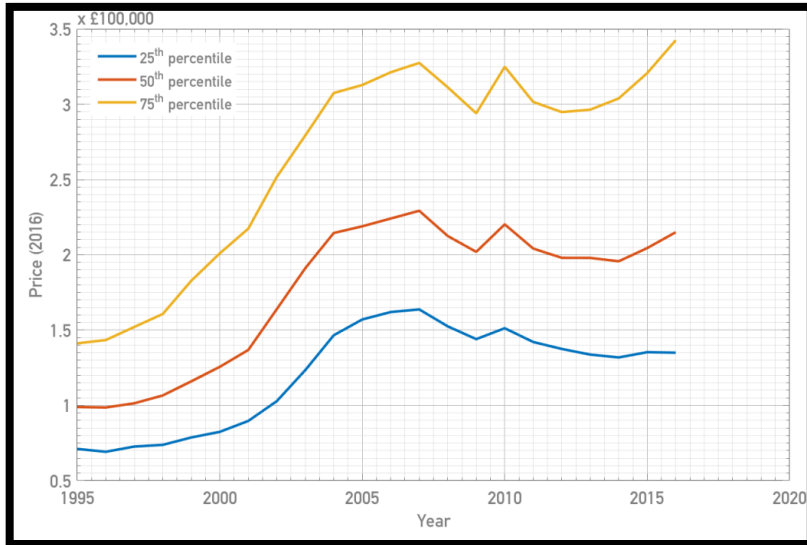
Why is it important to know when functions are increasing and decreasing?

Why should we care about maxima and minima?

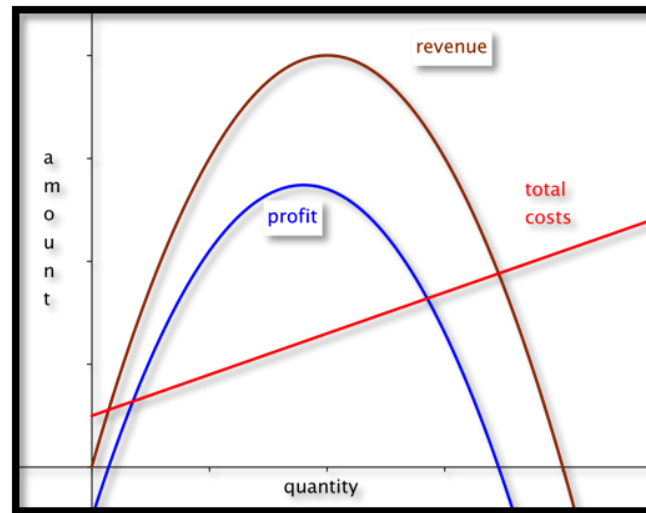


Short Answer

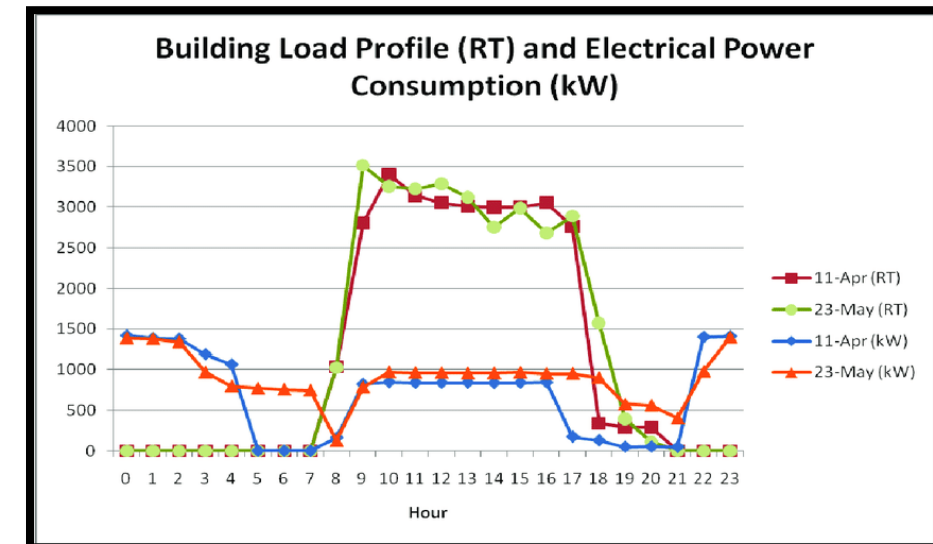
Why are maxima and minima important



House prices across the years

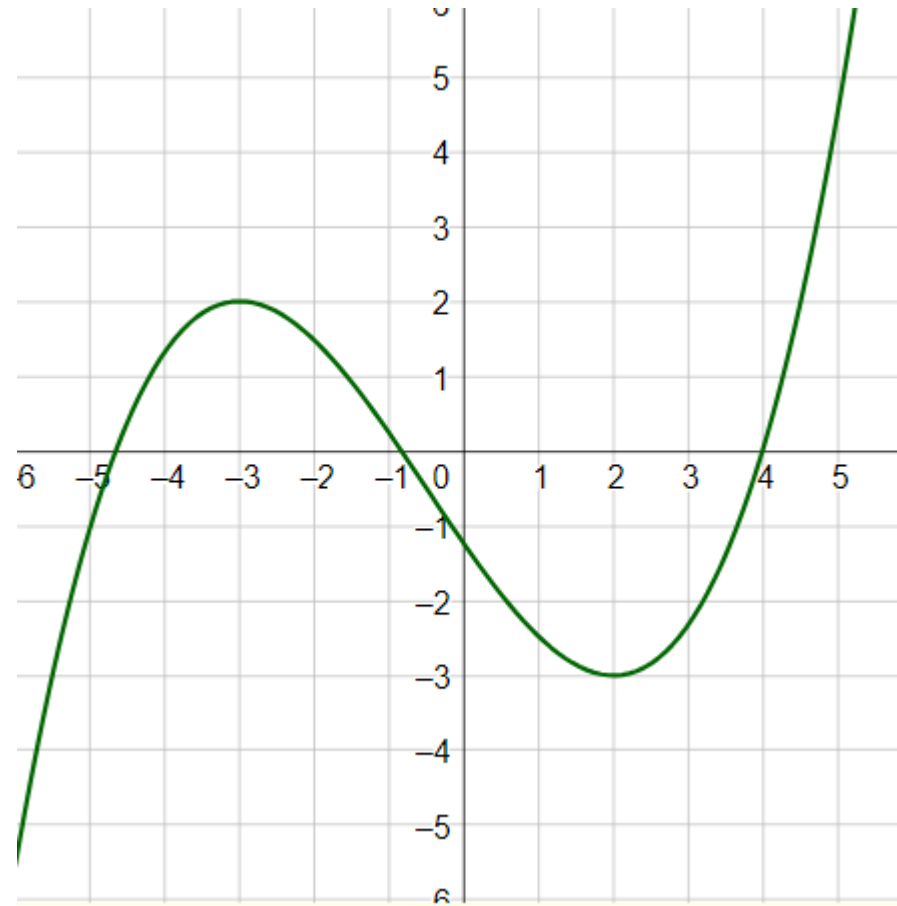


Modeling revenue, cost, profit



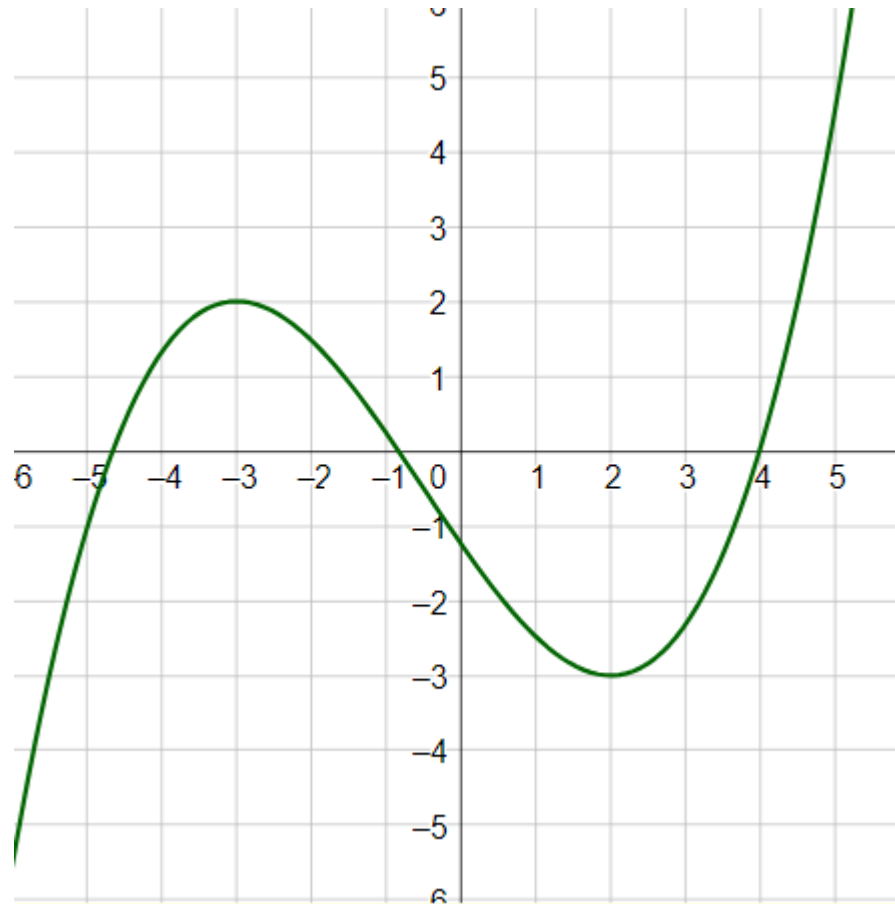
Electric power consumption

Find Local extrema from graph



EXAMPLE

Find Local extrema from graph



Solution

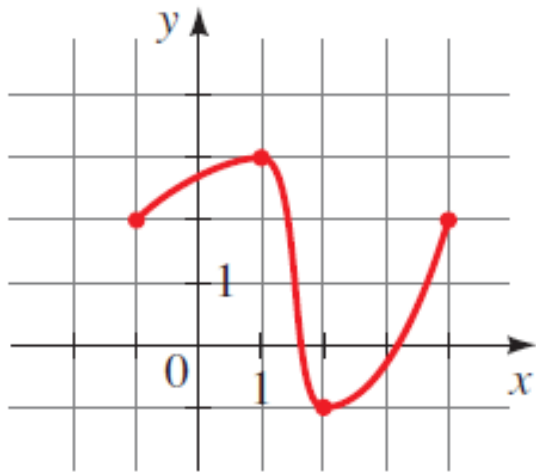
The maximums are the highest points in the area. For this graph there is only one maximum and it is $(-3, 2)$.

The minimums are the lowest points in the area. For this graph there is only one minimum and it is $(2, -3)$.

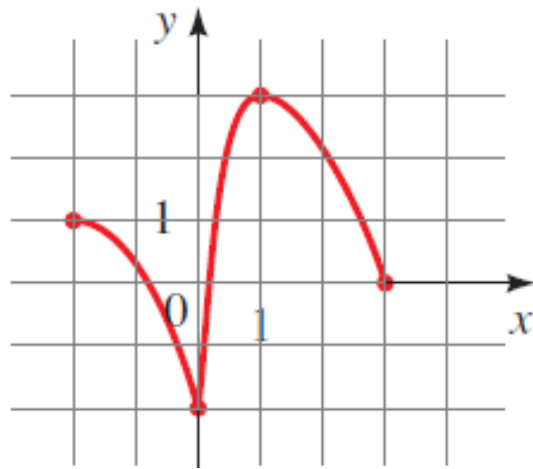
Increasing and Decreasing Functions

The graph of a function f is given. Use the graph to estimate

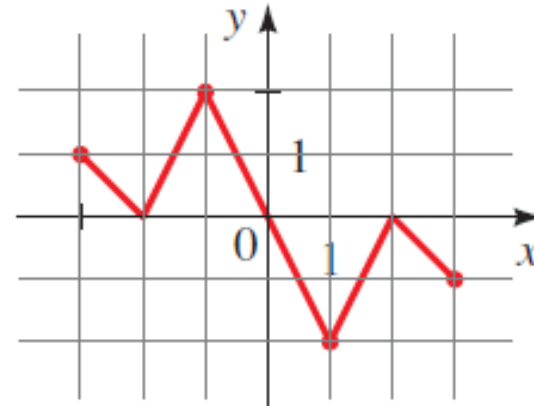
- (i) The domain and range of f .
- (ii) The intervals on which f is increasing and on which f is decreasing.



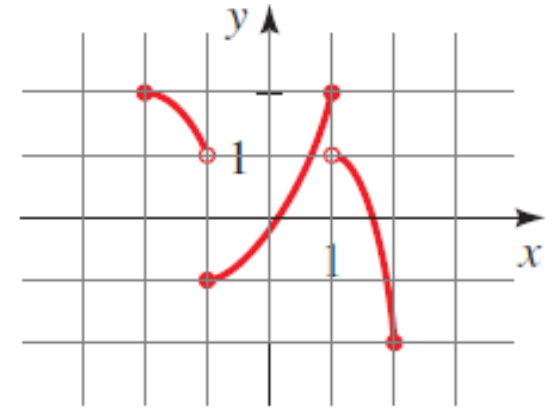
(a)



(b)



(c)



(d)

ACTIVITY 2.5

2.5 Increasing and Decreasing Functions

2.5 Increasing and Decreasing Functions

SOLUTIONS

2.5 Increasing and Decreasing Functions

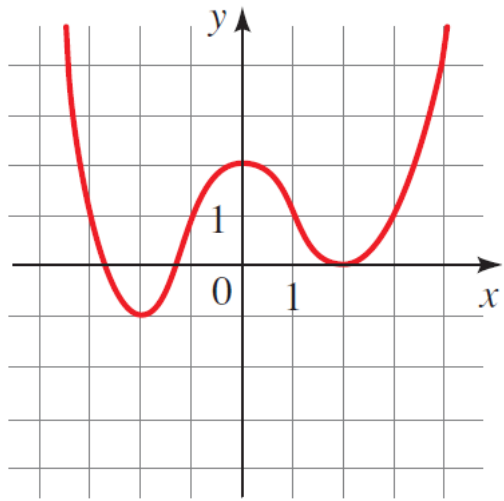
SOLUTIONS

2.5 Increasing and Decreasing Functions

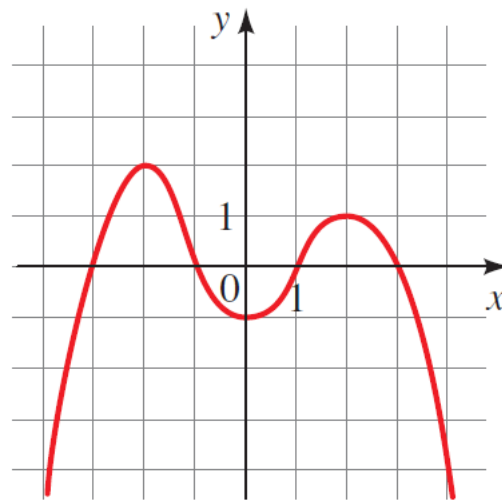
SOLUTIONS

Local Maximum and Minimum Values

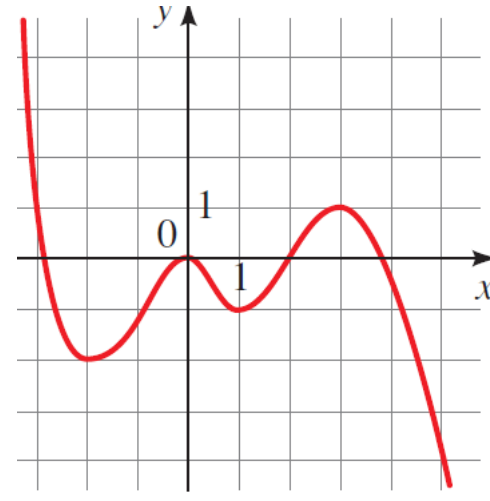
The graph of a function f is given. Use the graph to estimate all the local maximum and minimum values of the function and the value of x at which each occurs.



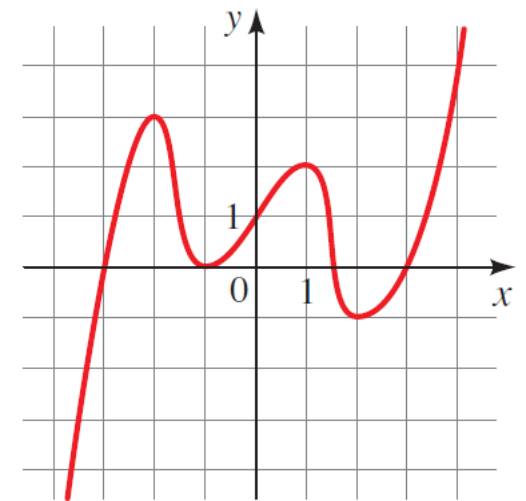
(a)



(b)



(c)



(d)

ACTIVITY 2.6

2.6 a, b: Local Maximum and Minimum Values

SOLUTION

2.6 c, d: Local Maximum and Minimum Values

SOLUTION